length—a millionth of a billionth of a billionth of a billionth (10^{-33}) of a centimeter—space becomes a seething, boiling cauldron of frenzied fluctuations. As the illustration makes clear, the usual notions of left/right, back/forth, and up/down become so jumbled by the ultramicroscopic tumult that they lose all meaning. Even the usual notion of before/after, which we've been illustrating by sequential slices in the spacetime loaf, is rendered meaningless by quantum fluctuations on time scales shorter than the *Planck time*, about a tenth of a millionth of a trillionth of a trillionth of a trillionth (10^{-43}) of a second (which is roughly the time it takes light to travel a Planck length). Like a blurry photograph, the wild undulations in Figure 12.2 make it impossible to distinguish one time slice from another unambiguously when the time interval between them becomes shorter than the Planck time. The upshot is that on scales shorter than Planck distances and durations, quantum uncertainty renders the fabric of the cosmos so twisted and distorted that the usual conceptions of space and time are no longer applicable.



Figure 12.2 Successive magnifications of space reveal that below the Planck length, space becomes unrecognizably tumultuous due to quantum jitters. (These are imaginary magnifying glasses, each of which magnifies between 10 million and 100 million times.)